★ AIxBio Hackathon 2026

Toxin Circuits in ESM-2

Mechanistic Interpretability Reveals Why Structure-Aware Probes Resist ProteinMPNN Redesign

Shivam Dubey , Manan Wadhwa

93.9%

ESM-2 Detection Rate vs 0% BLAST

0/1,179

Mutations Evade Detection

248

Zero-Shot Threat Discoveries

THE PROBLEM

Current Biosecurity Screening Is Broken

BLAST-Based Screening

- Relies on sequence identity thresholds (30–50%)
- ProteinMPNN redesigns toxin sequences below every threshold
- All 534 redesigns achieve < 30% identity to training toxins
- Detection rate: 0.0% across all thresholds
- Sequence-based screening is fundamentally insufficient

The Adversary's Toolkit

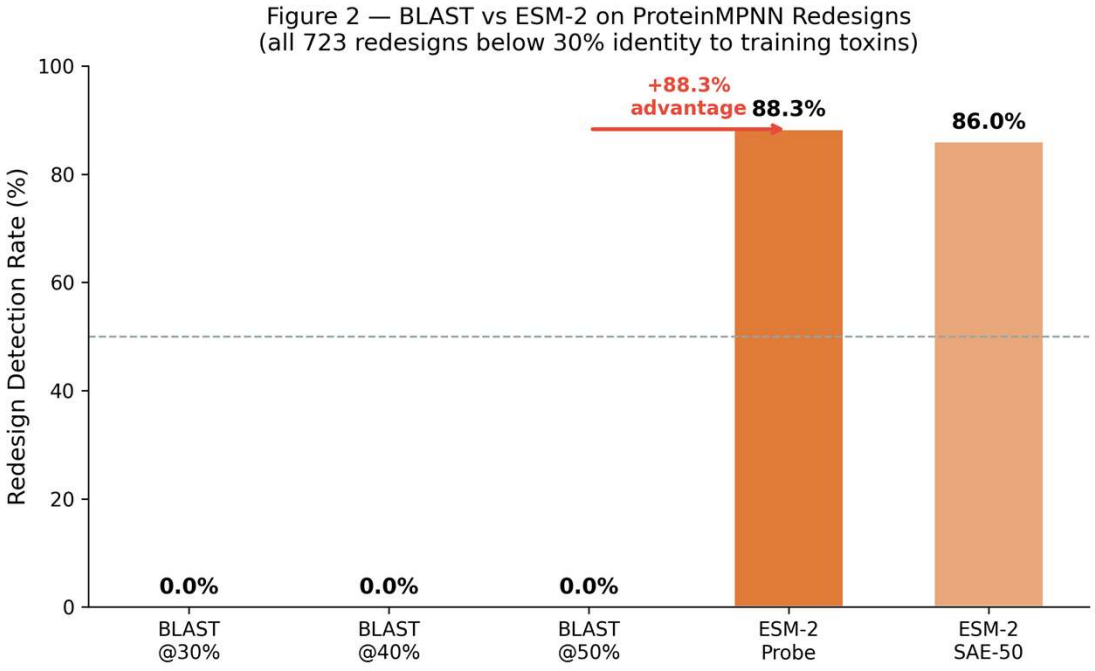
- ProteinMPNN: Structure-conditioned sequence redesign
- Preserves protein backbone geometry (fold topology)
- Generates novel sequences with same toxic function
- No access to screening models needed
- Creates proteins synthesizable via commercial providers

Question: Can protein language model probes detect what BLAST cannot?

HEADLINE RESULT

BLAST: 0% vs ESM-2: 93.9%

The ESM-2 probe detects what BLAST completely misses , with zero retraining



Headline Numbers

BLAST @ 30% identity: 0.0%

BLAST @ 40% identity: 0.0%

BLAST @ 50% identity: 0.0%

ESM-2 Probe (L33): 93.9%

ESM-2 SAE Top-50: 86.0%

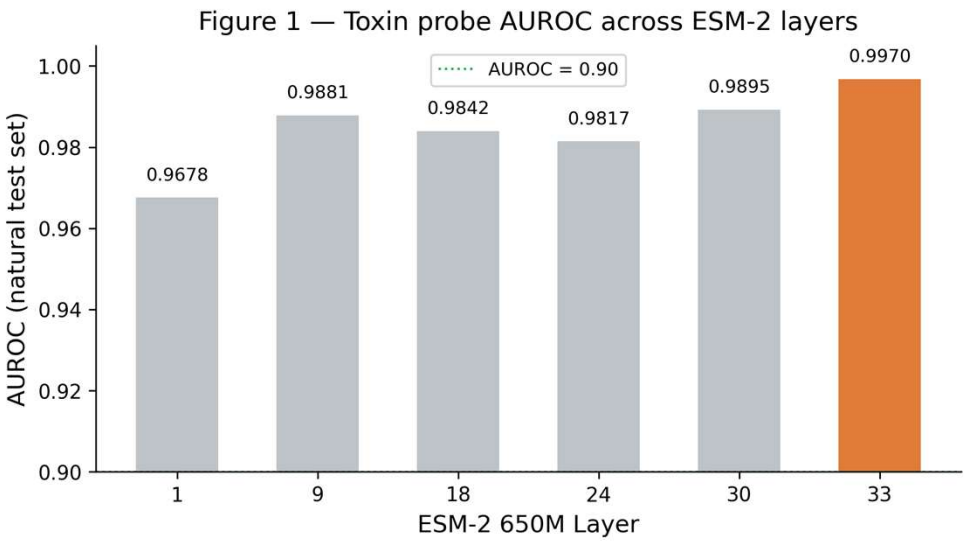
McNemar $p = 1.53 \times 10^{-63}$

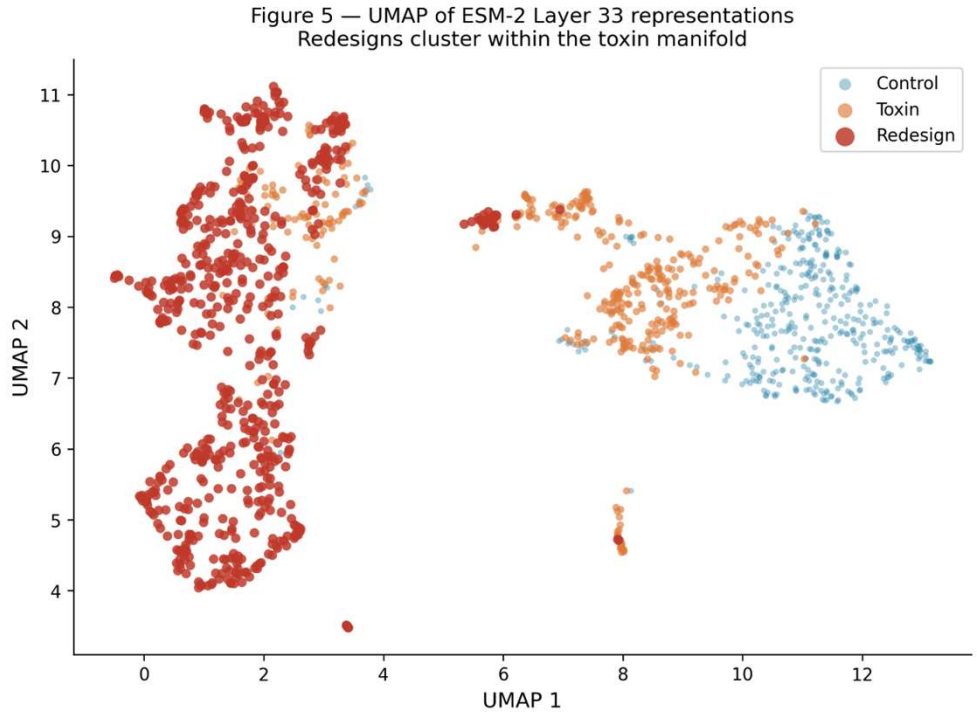
Pure generalisation , probe trained only on natural toxins & controls

PROBE ARCHITECTURE

Layer 33: The Toxin Bottleneck

Toxin signal builds progressively through ESM-2 , Layer 33 achieves AUROC 0.9970



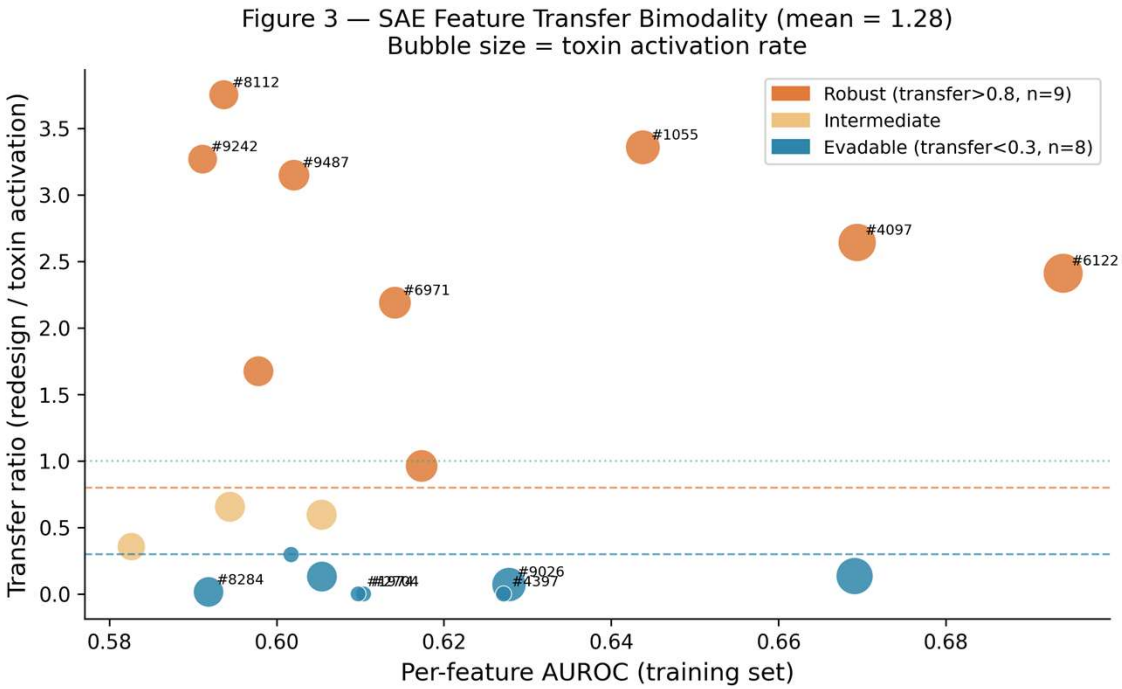


Signal is structural, not just compositional

Redesigns (red) cluster with toxins (orange), not controls (blue)

SAE CIRCUIT ANALYSIS

50 Sparse Features Explain the Probe at 205× Compression



Probe detection is a collective property , superposition hypothesis

1.28

Mean Transfer Ratio

205×

Compression Ratio

Bimodal Feature Landscape

Robust features (transfer > 1.0):

- Structural rigidity, Pro/Phe enriched
- Amplified by ProteinMPNN redesign

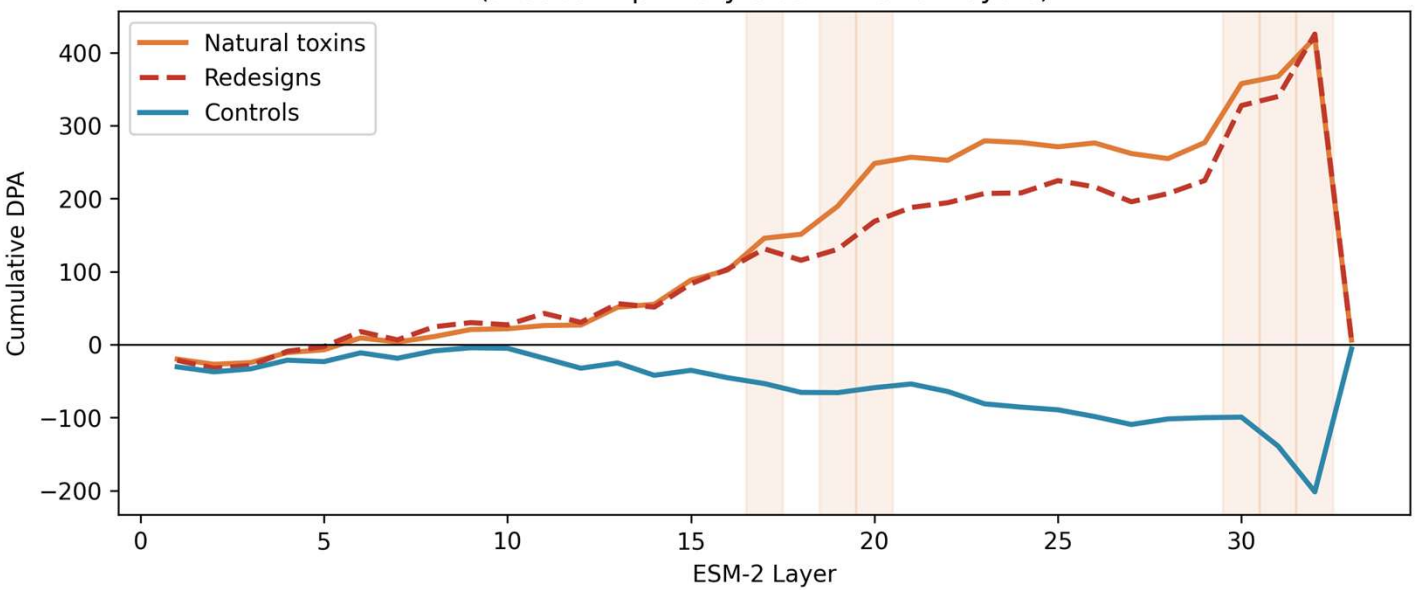
Evadable features (transfer < 0.2):

- Sequence-specific Cys patterns
- Collapsed by redesign

MECHANISTIC INTERPRETABILITY

Direct Probe Attribution: Layer 32 Is the Bottleneck

Figure 6 — Toxin signal accumulation through ESM-2 depth
(shaded = primary discrimination layers)



The fold constrains the endpoint,

not the route through the network

Circuit Architecture

Layer 32 DPA gap: +130.1

→ Primary circuit bottleneck

Tox/Redesign correlation: r = 0.992

→ Same representational pathway

Circuit overlap: only 19%

→ Multiple routes to toxin endpoint

160 attention heads ablated:

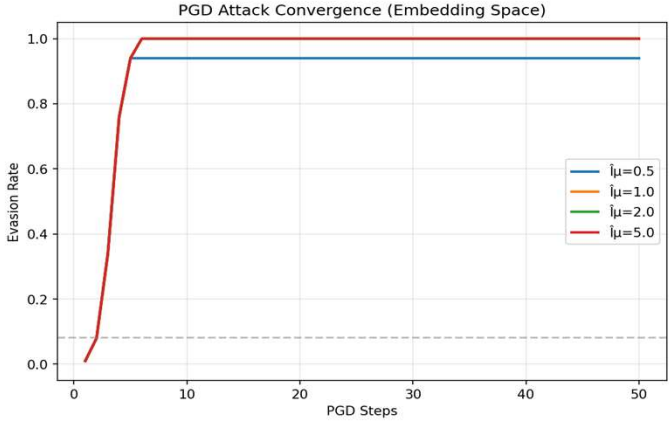
→ Distributed, redundant processing

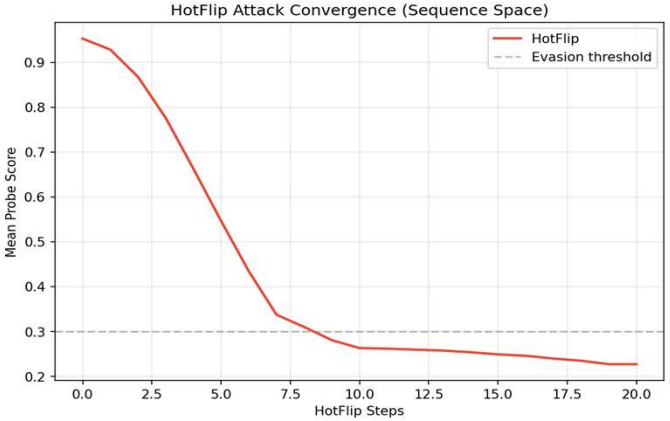
→ No single head dominates

ADVERSARIAL ROBUSTNESS ★ NOVEL FRAMING

Four-Tier Attack Taxonomy: Security = Weight Privacy

ProteinMPNN	HotFlip	PGD ε=2	pSSR / UAP
Analogy: Blackbox	Analogy: GCG analogue	Analogy: PGD	Analogy: Oracle
Space: Sequence	Space: Seq (gradient)	Space: Embedding	Space: Embedding
Access: Structure only	Access: ESM-2 + probe	Access: Full whitebox	Access: Full whitebox
6.1%	**100%†**	**100%**	**100%**

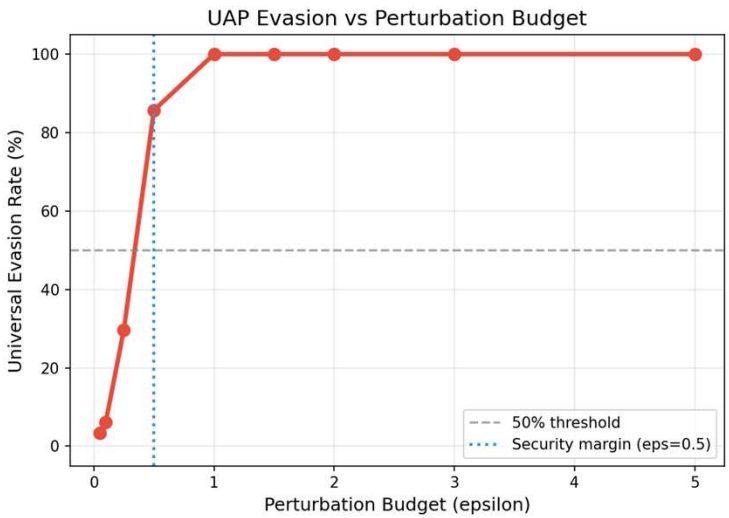


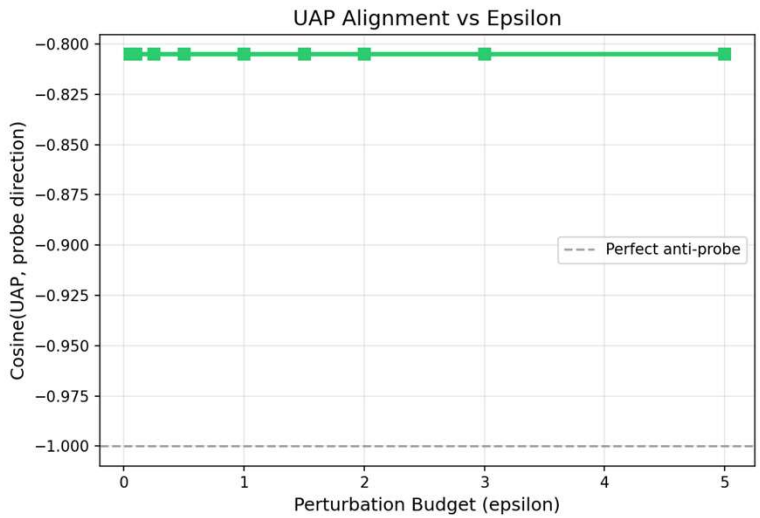


Security boundary lies at gradient access not sequence vs. embedding space

UAP MANIFOLD GEOMETRY ★ NOVEL ANALYSIS

A Single Vector Breaks Every Toxin , But Only With Gradient Access





Probe security = weight privacy

Key Findings

Security margin at ε = 0.5
→ Evasion first crosses 50%

cos(UAP, probe) = −0.805
→ Invariant across ALL ε values
→ Structural property of manifold

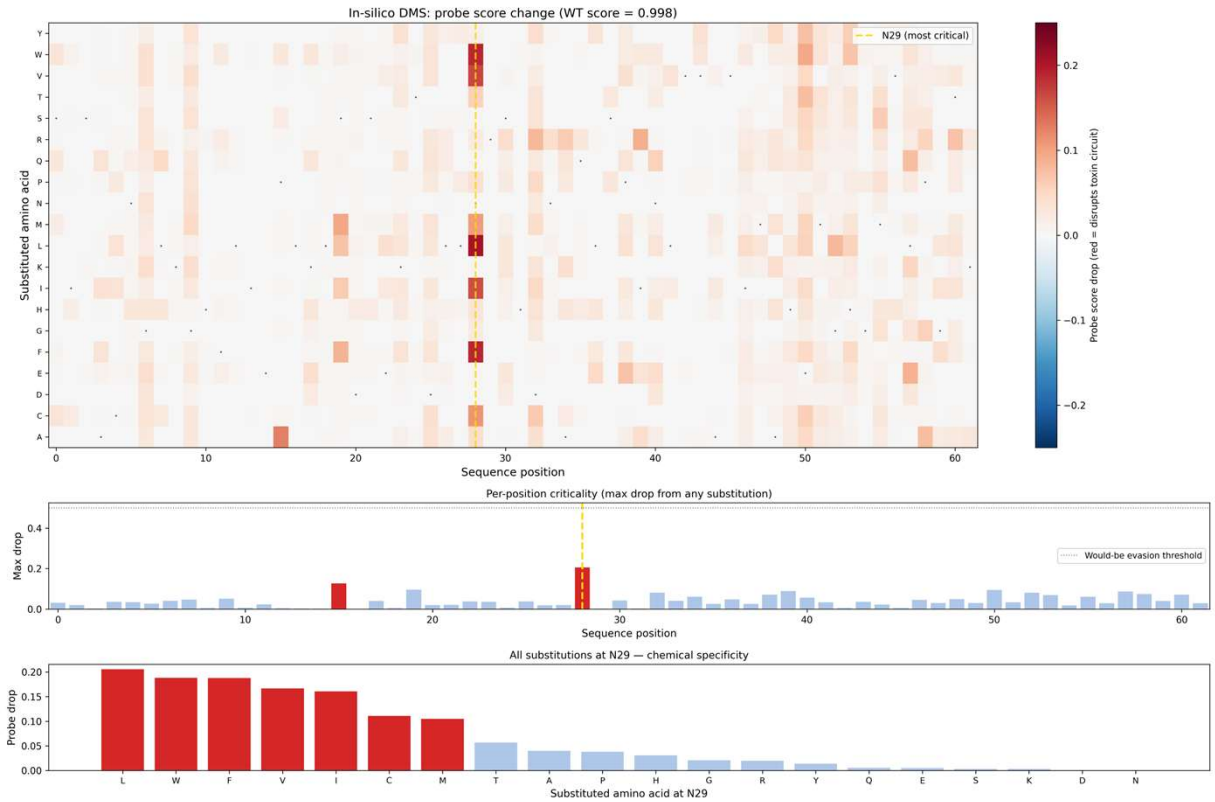
UAP deviates 36° from anti-probe
→ Not simply the inverse direction

Suppresses robust SAE features
#4097: Δ = −0.048
#1055: Δ = −0.045

MUTATION ROBUSTNESS

0/1,179 Single-Point Mutations Evade Detection

The circuit reads chemical polarity, not amino acid identity



Position N29 Analysis

Polar substitutions (D, S, K, E, Q):
Score drop < 0.01
→ Invisible to circuit

Hydrophobic (I, V, F, W, L):
Score drop 0.16 – 0.21
→ Partial disruption only

Even N29G (no sidechain):
Drop = 0.021 only

Circuit reads hydrogen-bonding network of the toxin fold

ZERO-SHOT THREAT DISCOVERY ,CROSS-KINGDOM

Zero Shot Discovery: The probed learned the secrets

Trained on animal & bacterial toxins only , discovers threats across kingdoms, folds, and annotation gaps

248 / 1,000
UniRef50 flagged (>0.85)

4.75×
Signal peptide enrichment

54.4%
Currently uncharacterised

p < 0.001
Fisher's exact test

0.9905
Ecp2 fungal effector score

Hit	Organism	Score
Ecp2 effector protein*	Colletotrichum (fungus)	**0.9905**
Hemolysin, chromosomal	Candidatus Accumulibacter	**0.9996**
Leukotoxin	Candidatus Accumulibacter	**0.9922**
Uncharacterised protein†	A. baumannii (WHO P1)	**0.9995**
GDSL-like Lipase‡	A. baumannii 625974 (WHO P1)	**0.9993**
Cyclolysin	Candidatus Accumulibacter	**0.9810**

* Cross-kingdom: trained on no fungi † Confirmed GDSL-like Lipase via AlphaFold (pLDDT=83.0) ‡ Secreted membrane-degrading enzyme

Three Lines of Evidence

1. Cross-Kingdom Transfer

Ecp2 fungal effector detected despite zero fungi in training. 23 total fungal hits from Colletotrichum/Pezizomycotina.

2. Structure-Agnostic Detection

50/50 split: well-folded (pLDDT≥70) and disordered proteins both scored >0.99. Reads sequence motifs, not fold.

3. Signal Peptide Enrichment

4.75x enrichment (38% vs 8%) confirms the learned concept is "secreted" — the defining property of toxins & effectors.

SAE Recovers 38% of "Double-Evaders" Proving Disentanglement

Sequences that fool both BLAST and the linear probe still contain toxin-relevant features in sparse representation

McNemar Contingency (n = 534)

	Probe+	Probe−
BLAST+	107	13
BLAST−	**322**	**92**

Probe-only: 322 (60%) | Double-evaders: 92 (17%)

$p = 1.53 \times 10^{-63}$

SAE Recovery of Double-Evaders

34%
LogReg on SAE top-50

38%
ToxinProbe on SAE top-50

SAE features are direction-sensitive

→ Detect active motifs regardless of global distance

Linear probes are distance-sensitive

→ Miss sequences outside training radius

ESM-2 internally represents toxicity even when the probe can't surface it

GENERALISATION

Zero-Shot Generalisation Across Toxin Families

Held-out Family	n	Train AUROC	Holdout AUROC
Neurotoxin	32	0.9991	0.9951
Phospholipase	39	0.9993	0.9987
Conotoxin	260	0.9990	0.9994
Snake toxin	22	0.9997	0.9783
Mean			**0.9929**

Activation Steering , Toxicity Is a Causal Linear Direction

Control baseline (lowest 50)	Steered at α = +2.0	Natural toxin baseline
0.000	1.000 ★	0.964

NOVELTY & IMPACT

What Makes This Novel vs Prior Work

This Paper

- Explains WHY probes work mechanistically
- SAE circuit analysis (50 features, 205× compression)
- Four-tier attack taxonomy with security boundary
- Zero-shot cross-kingdom discovery (fungi!)
- UAP manifold geometry analysis
- SAE disentanglement of blind spots (38% recovery)
- DMS polarity readout (0/1,179 evasion)

Wittmann et al. 2025

- Shows probes work empirically
- No mechanistic analysis
- Single attack type (ProteinMPNN)
- No discovery component
- No adversarial robustness analysis
- No blind spot analysis
- No sequence robustness analysis

CONCLUSIONS

Key Takeaways

BLAST: 0%. ESM-2: 93.9%.	ProteinMPNN fails to evade structure-aware probes
Transfer ratio 1.28	Redesigns amplify toxin features , adversary works against itself
r = 0.992	Redesigns use the same DPA circuit as natural toxins
Security = Weight Privacy	6.1% evasion (blackbox) vs 100% (whitebox gradient)
Zero-shot discovery	Cross-kingdom detection: Hemolysin, Leukotoxin, fungal Ecp2
0/1,179 mutations evade	Circuit reads chemical polarity, not sequence identity

ProteinMPNN cannot redesign away what is structurally necessary for toxin function , and ESM-2 reads structure.

FUTURE WORK

Open Problems & Next Steps

1. **HotFlip Transferability**
   Do gradient-guided attacks transfer across pLM architectures?

2. **Protein Design Alignment**
   Fine-tune ProteinMPNN with layer-32 interventions , protein RLHF analogue

3. **Wet-Lab Validation**
   Two A. baumannii candidates (0.9995, 0.9993) need cytotoxicity assays

4. **Multi-Generator Benchmark**
   Extension to RFdiffusion & EvoDiff , systematic cross-generator study

5. **Multi-Scale Probes**
   Combine global embedding distance with sparse SAE feature activation

Thank You

Shivam Dubey
Manan Wadhwa

AIxBio Hackathon 2026

Apart Research

Code & Notebooks:

https://github.com/punctualprocrastinator/cautious-goggles

Contact:
dubeyjiworks@gmail.com
wadhwamanan56@gmail.com